CS105.3Faculty of Computing, Online Examinations 2021

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MODULE CODE	**MA101.3**		
MODULE LECTURER	**Mr. Gayan Perera**	**DATE SUBMITTED**	**28/10/ 2021**

For office purpose only:

GRADE/MARK	
COMMENTS	

Declaration

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Date: 28/10/ 2021 …………………………… **E- Signature:

Question 1

①

a) $\frac{1}{27} = 0.75 \rightarrow$ Rational

b) $\sqrt{6} = 2.4494\ldots \rightarrow$ Irrational

c) $\frac{22}{7} = 3.1428\ldots \rightarrow$ Irrational

d) $1/3 = 0.333\ldots \rightarrow$ Rational

e) $2/\sqrt{3} = 1.154\ldots \rightarrow$ Irrational

②

a) $(p^2)^4 = p^8$

b) $(p^{-3})^5 = p^{-15} = \frac{1}{p^{15}}$

c) $(p^{-5})^4 = p^{-20} = \frac{1}{p^{20}}$

d) $y^3(y^{-5})^3 = y^3 . y^{-15}$

$= y^{-12} = \frac{1}{y^{12}}$

e) $(4m)^4 = 16m^4$

③

a) $4^x = 2^6$

$\rightarrow \quad 2^{2x} = 2^6$

$\therefore \quad 2x = 6$

$\underline{\underline{x = 3}}$

b) $3^{2x} = 9^4$

$\rightarrow \quad 3^{2x} = 3^8$

$\therefore \quad 2x = 8$

$\underline{\underline{x = 4}}$

c) $2^x = \frac{1}{32}$

$\Rightarrow \quad 2^x = \frac{1}{2^5}$

$\Rightarrow \quad 2^x = 2^{-5}$

$\Rightarrow \quad \underline{\underline{x = -5}}$

4

a) $\frac{y^5}{y^4} = y$

b) $\frac{d^3}{d^9} = d^{-6}$

c) $\frac{x^4 y^{-1} z^8}{z x^4 y^5} = \frac{\cancel{x^4} y^{-1} z^8}{z \cancel{x^4} y^5}$

$= y^{-6} z^7$

(5)

a) $3^2 (3x)^3$

$= 3^2 . 27 x^3$

$= \cancel{81 x^3}$ $243 x^3$

b) $(4.1)^5 (4.1)^{-5}$

$= \frac{\cancel{(4.1)^5}}{\cancel{(4.1)^5}}$

$= 1$

c) $(b^5)^3 (b^2)$

$= b^{15} b^2$

$= b^{17}$

Question 2

1

a) $0 \notin \{2, 4, 5, 6\}$

b) $\{1, 4\} \subseteq \{2, \{1, 4\}, 7\}$

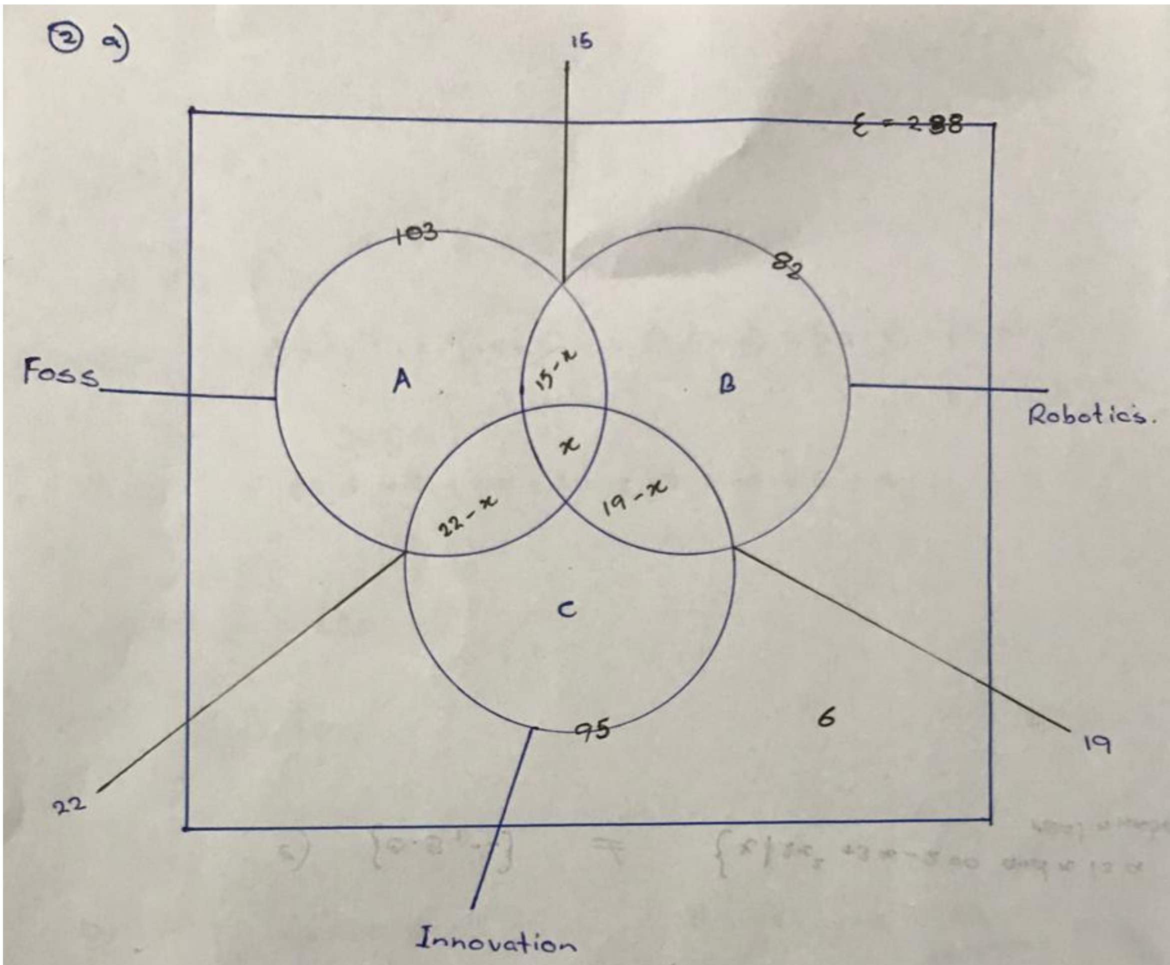
~~c) $\{2, 3\}$ $\{2, \{1, 4\}, 7\}$~~

c) $\{2, 3\} \subseteq \{1, 2, 3, 4\}$

d) $4 \in \{1, \{2, 3\}, 4, 5\}$

e) $\{0.5, -2\} \neq \{x \mid 2x^2 + 3x - 2 = 0 \text{ and } x \text{ is a real number}\}$

② a)



b)

$A = 103 - (22-x + 15-\cancel{x} + \cancel{x})$

$= 103 - 22 + x - 15$

$= 66 + x$

$B = 82 - (15-x + 19-\cancel{x} + \cancel{x})$

$= 82 - 15 + x - 19$

$= 48 + x$

$C = 95 - (22-x + 19-\cancel{x} + \cancel{x})$

$= 95 - 22 + x - 19$

$= 54 + x$

~~28~~

$$238 = A + B + C + 6 + 22 - x + 19 - x + 15 - x + x$$

$$238 = (66 + \cancel{x}) + (48 + \cancel{x}) + (54 + \cancel{x}) + (22 - \cancel{x}) + (19 - \cancel{x}) + (15 - \cancel{x}) + x + 6$$

$$238 = 66 + 48 + 54 + 22 + 19 + 15 + 6 + x$$

$$238 = 230 + x$$

$$x = 8$$

b) Only members of Innovation club $= 54 + x \rightarrow 62$

c) Students are in all three clubs $\rightarrow 8$

③

a) $A \times (B \cup C) = \{(a,2),(b,2),(a,3),(b,3),(a,5),(b,5)\}$

$\boxed{(B \cup C) = \{2,3,5\}}$

b) $(A \times B) \cup (A \times C) = \{(a,2),(b,2),(a,3),(b,3),(a,5),(b,5)\}$

* $(A \times B) = (a,3),(b,3),(a,5),(b,5)$

* $(A \times C) = (a,2),(b,2),(a,3),(b,3)$

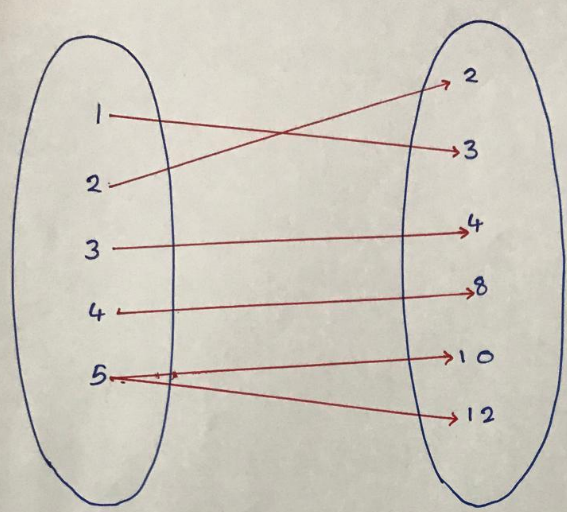
c) $A \times (B \cap C) = \{(a,3),(b,3)\}$

* $B \cap C = \{3\}$

d) $(A \times B) \cap (A \times C) = \{(a,3),(b,3)\}$

4

a)



Domain

(A set of all x's)

Range

(B set of all y's)

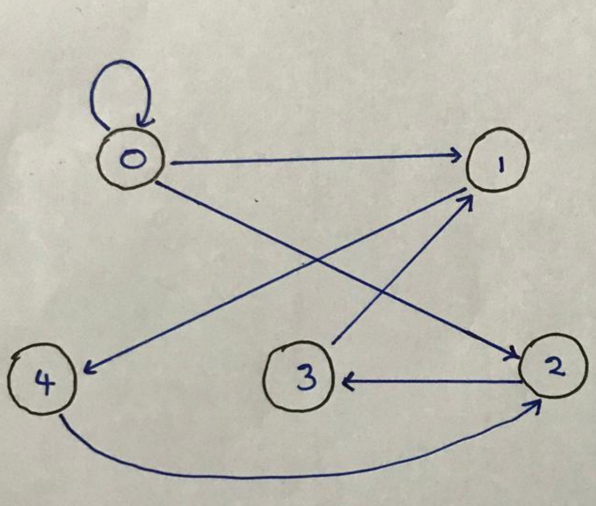
b) No

→ ∵ The element '5' is paired two elements. of the range. So it is not a Function.

5

$A = \{0, 1, 2, 3, 4\}$

Let $R = \{(0,0), (0,1), (0,2), (1,4), (2,3), (3,1), (4,2)\}$



Question

② (a)

p	q	r	$\sim p$	$\sim q$	$\sim p \vee \sim q$	$\sim p \vee \sim q \wedge r$
F	F	F	T	T	T	F
F	F	T	T	T	T	T
F	T	F	T	F	T	F
F	T	T	T	F	T	T
T	F	F	F	T	T	F
T	F	T	F	T	T	T
T	T	F	F	F	F	F
T	T	T	F	F	F	T

(b)

p	q	r	$\sim p$	$\sim r$	$(\sim p) \vee (q)$	$(\sim p) \vee (q) \wedge (\sim r)$
F	F	F	T	T	T	T
F	F	T	T	F	T	F
F	T	F	T	T	T	T
F	T	T	T	F	T	F
T	F	F	F	T	F	F
T	F	T	F	F	F	F
T	T	F	F	T	T	T
T	T	T	F	F	T	F

③

a)

p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$(\sim p) \vee (\sim q)$
F	F	F	T	T	T	T
F	T	F	T	T	F	T
T	F	F	T	F	T	T
T	T	T	F	F	F	F

$\downarrow$ L.H.S $\qquad$ $\downarrow$ R.H.S.

$\therefore$ L.H.S. = R.H.S

b

p	q	$p \vee q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$(\sim p) \wedge (\sim q)$
F	P	F	T	T	T	T
F	T	T	F	T	F	F
T	F	T	F	F	T	F
T	T	T	F	F	F	F

$\downarrow$ L.H.S $\qquad$ $\downarrow$ R.H.S.

= L.H.S = R.H.S

c

p	q	r	$p \vee q$	$(p \vee q) \wedge r$	$\sim\{(p \vee q) \wedge r\}$	$\sim p$	$\sim q$	$\sim r$	$\sim p \vee \sim q$	$(\sim p \wedge \sim q) \vee \sim r$
F	F	F	F	F	T	T	T	T	T	T
F	F	T	F	F	T	T	T	F	T	T
F	T	F	T	F	T	T	F	T	F	T
F	T	T	T	T	F	T	F	F	F	F
T	F	F	T	F	T	F	T	T	F	T
T	F	T	T	T	F	F	T	F	F	F
T	T	F	T	F	T	F	F	T	F	T
T	T	T	T	T	F	F	F	F	F	F
					↓ L.H.S					↓ R.H.S

L.H.S = R.H.S.

4 (a)

p	q	$(p \Rightarrow q)$	$(q \Rightarrow p)$	$(p \Rightarrow q \wedge q \Rightarrow p)$
F	F	T	T	T
F	T	T	F	F
T	F	F	T	F
T	T	T	T	T

No :   Date: / /

b)

p	q	$p \wedge q$	$\sim(p \wedge q)$	$q \Rightarrow p$	$\sim q \Leftrightarrow p$	[illegible]
F	F	F	T	T	F	T
F	T	F	T	F	T	T
T	F	F	T	F	T	T
T	T	T	F	T	F	F

Question 4

①

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 8 & 3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 0 & 1 \end{bmatrix}$$

a) $A + B$ ⇒ ~~2,3~~ The Order of the matrix A is 2×3 and the order of the matrix B is 3×2. So that is not equal.

or

b) $2A - 3b$

we can't do addition or substraction for these two.

c)

$$A \times B = \begin{bmatrix} 3 & 2 & 1 \\ 8 & 3 & 4 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 0 & 1 \end{bmatrix}_{3 \times 2}$$

$$= \begin{bmatrix} 3-4 & -3+4+1 \\ 8-6 & -8+6+4 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} -1 & 2 \\ 2 & 2 \end{bmatrix}$$

(2)

i) $A = \begin{bmatrix} -3 & -5 \\ 2 & 6 \end{bmatrix}$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 6 & 5 \\ -2 & -3 \end{bmatrix}$$

$$= \frac{1}{-8} \begin{bmatrix} 6 & 5 \\ -2 & -3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -3/4 & -5/8 \\ 1/4 & 3/8 \end{bmatrix}$$

ii) $B = \begin{bmatrix} 7 & 3 \\ -2 & -1 \end{bmatrix} \rightarrow B^{-1} = \frac{1}{\det|B|} \begin{bmatrix} -1 & -3 \\ +2 & 7 \end{bmatrix}$

$$= \frac{1}{-13} \begin{bmatrix} -1 & -3 \\ +2 & 7 \end{bmatrix}$$

$$B^{-1} = \begin{bmatrix} 1/13 & 3/13 \\ -2/13 & -7/13 \end{bmatrix}$$

(3)

$3x + 4y = 25$

$4x + 3y = 24$

$$\begin{bmatrix} 3 & 4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 25 \\ 24 \end{bmatrix}$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

$$A \qquad x \qquad C$$

$$A^{-1} = \frac{1}{9-16} \begin{bmatrix} 3 & -4 \\ -4 & 3 \end{bmatrix}$$

$$= \frac{1}{-7} \begin{bmatrix} 3 & -4 \\ -4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -3/7 & 4/7 \\ 4/7 & -3/7 \end{bmatrix}$$

$$AX = C$$

$$X = A^{-1}C$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3/7 & 4/7 \\ 4/7 & -3/7 \end{bmatrix} \begin{bmatrix} 25 \\ 24 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -75/7 + 96/7 \\ 100/7 - 72/7 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$\therefore x = 3$

$y = 4$

4

$$A^{-1} = \frac{(\text{Co. matrix})^T}{|A|} \qquad A = \begin{bmatrix} 1 & 7 & -6 \\ 8 & 4 & 3 \\ -6 & -2 & 5 \end{bmatrix}$$

$$C.M = \begin{bmatrix} \begin{vmatrix} 4 & 3 \\ -2 & 5 \end{vmatrix} & -\begin{vmatrix} 8 & 3 \\ -6 & 5 \end{vmatrix} & \begin{vmatrix} 8 & 4 \\ -6 & -2 \end{vmatrix} \\ -\begin{vmatrix} 7 & -6 \\ -2 & 5 \end{vmatrix} & \begin{vmatrix} 1 & -6 \\ -6 & 5 \end{vmatrix} & -\begin{vmatrix} 1 & 7 \\ -6 & -2 \end{vmatrix} \\ \begin{vmatrix} 7 & -6 \\ 4 & 3 \end{vmatrix} & -\begin{vmatrix} 1 & -6 \\ 8 & 3 \end{vmatrix} & \begin{vmatrix} 1 & 7 \\ 8 & 4 \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} (20+6) & -(40+18) & (-16+24) \\ -(35-12) & (5-36) & -(-2+42) \\ 21+24 & -(3+48) & (4-56) \end{bmatrix}$$

$$C.M. = \begin{bmatrix} 26 & -58 & 8 \\ -23 & -31 & -40 \\ 45 & -51 & -52 \end{bmatrix}$$

$$C.M^T = \begin{bmatrix} 26 & -23 & 45 \\ -58 & -31 & -51 \\ 8 & -40 & -52 \end{bmatrix}$$

$$|A| = 1\begin{bmatrix} 4 & 3 \\ -2 & 5 \end{bmatrix} - 7\begin{bmatrix} 8 & 3 \\ -6 & 5 \end{bmatrix} \cancel{+} 6\begin{bmatrix} 8 & 4 \\ -6 & -2 \end{bmatrix}$$

$$|A| = 1(20+6) - 7(40+18) - 6(-16+24)$$

$$|A| = 26 - 406 - 48$$

$$|A| = 428$$

$$\therefore \quad A^{-1} = \frac{C.M^T}{|A|}$$

$$= \frac{1}{428}\begin{bmatrix} 26 & -23 & 45 \\ -58 & -31 & -51 \\ 8 & -40 & -52 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 0.06 & -0.053 & 0.105 \\ -0.135 & -0.072 & -0.119 \\ 0.018 & -0.093 & 0.121 \end{bmatrix}$$

R = RAMS

V = VGA

S = SSD

(5)

$$2R + 3V + 4S = 38000$$

$$4R + 2V + 3S = 32000$$

$$3R + 4V + 2S = 38000$$

$$\underbrace{\begin{bmatrix} 2 & 3 & 4 \\ 4 & 2 & 3 \\ 3 & 4 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} R \\ V \\ S \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 38000 \\ 32000 \\ 38000 \end{bmatrix}}_{C}$$

$$x = A^{-1}C$$

$$\rightarrow * \; A^{-1} = \frac{C.M^T}{|A|}$$

$$C.M = \begin{bmatrix} \begin{vmatrix} 2 & 3 \\ 4 & 2 \end{vmatrix} & -\begin{vmatrix} 4 & 3 \\ 3 & 2 \end{vmatrix} & \begin{vmatrix} 4 & 2 \\ 3 & 4 \end{vmatrix} \\ -\begin{vmatrix} 3 & 4 \\ 4 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 4 \\ 3 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} \\ \begin{vmatrix} 3 & 4 \\ 2 & 3 \end{vmatrix} & -\begin{vmatrix} 2 & 4 \\ 4 & 3 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ 4 & 2 \end{vmatrix} \end{bmatrix}$$

$$C.M = \begin{bmatrix} (4-12) & -(8-9) & (16-6) \\ -(6-16) & (4-12) & -(8-9) \\ (9-8) & -(6-16) & (4-12) \end{bmatrix}$$

$$C.M. = \begin{bmatrix} -8 & 1 & 10 \\ 10 & -8 & 1 \\ 1 & 10 & -8 \end{bmatrix}$$

$$C.M^{T} = \begin{bmatrix} -8 & 10 & 1 \\ 1 & -8 & 10 \\ 10 & 1 & -8 \end{bmatrix}$$

$$|A| = 2\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix} - 3\begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix} + 4\begin{bmatrix} 4 & 2 \\ 3 & 4 \end{bmatrix}$$

$$= 2(4-12) - 3(8-9) + 4(16-6)$$

$$= -16 + 3 + 40$$

$$= \underline{\underline{27}}$$

$$\therefore \quad A^{-1} = \frac{C.M^T}{|A|}$$

$$A^{-1} = \frac{1}{27}\begin{bmatrix} -8 & 10 & 1 \\ 1 & -8 & 10 \\ 10 & 1 & -8 \end{bmatrix}$$

$$x = A^{-1} C$$

$$x = \frac{1}{27}\begin{bmatrix} -8 & 10 & 1 \\ 1 & -8 & 10 \\ 10 & 1 & -8 \end{bmatrix}\begin{bmatrix} 38000 \\ 32000 \\ 38000 \end{bmatrix}$$

Let's assume,

38000 = 38

32000 = 32

$$x = \begin{bmatrix} -304 + 320 + 38 \\ 38 \cancel{+} 256 + 380 \\ 380 + 32 - 304 \end{bmatrix}$$

$$\begin{bmatrix} R \\ V \\ S \end{bmatrix} = \begin{bmatrix} 54000 \\ 162000 \\ 108000 \end{bmatrix}$$

RAM = Rs. 54000

VGA = Rs. 162000

SSD = Rs. 108000.

Question 5

5

a)

$$2y = -5x + 7$$

$$y = mx + c$$

$$\frac{2y}{2} = \frac{-5x}{2} + \frac{7}{2}$$

$$y = -\frac{5x}{2} + \frac{7}{2}$$

$$m = \frac{-5}{2} \qquad c = 7/2$$

b)

$$y - 3x = -5$$

$$y = 3x + c$$

$$y = +3x - 5$$

$$m = 3 \qquad c = -5$$

(c)

$$y = 4$$

$$y = x + 4$$

$$m = 1 \qquad c = 4$$